Chiao-Tung (Carol) Chan

Staff Software Engineer / R&D Manager, Autonomous Systems

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Professional Summary

ISO 26262 Certified (AFSP) R&D Manager and Autonomous Systems Architect with 7+ years of full-stack experience spanning algorithm design (Localization & Tracking) to low-level hardware optimization (Edge AI & Compute). Proven track record of leading cross-functional teams to build 0-to-1 autonomous driving systems, deploying robust Visual-Inertial Localization and Sensor Fusion pipelines in international testing fields. Deep expertise in C++ system architecture, mitigating complex HW/SW bottlenecks (zero-copy memory management, TensorRT acceleration), and establishing scalable, safety-critical CI/CD infrastructure.

Experience

Industrial Technology Research Institute (ITRI) — **Hsinchu, Taiwan**
R&D Manager, Self-Driving Vehicle Department — *Apr 2019 – Jun 2026*

1. Algorithm Development: Localization & Tracking

- Spearheaded the 0-to-1 architecture and cross-border deployment of a safety-critical Visual-Inertial Localization pipeline (Camera/IMU/Odometry), establishing RTK-backed ground-truth validation infrastructure. Formulated production-grade state-estimation strategies under severe edge-device compute constraints, successfully driving international commercialization at the Toyota Driving School (Japan) and Taiwan testing fields within a tight 12-month window.
- Defined the enterprise technical vision for a modular C++ Object-Tracking Framework. Architected extensible abstract interfaces for heterogeneous sensor modalities (Radar x/y/z/rv, Camera IDs) utilizing Fixed-Lag IMM-RAUKF algorithms, creating a 2x development-velocity multiplier that unblocked parallel delivery for both Localization and Fusion engineering teams.
- Innovated proprietary modifications to traditional BEV-based Iterative Closest Point (ICP) algorithms, extending its capability to achieve precise 3D object detection natively within the visual-inertial localization pipeline. This algorithmic breakthrough, alongside a novel localization-based dynamic camera calibration method, successfully resulted in two commercially valuable patents (one Sole Inventor, one Lead Inventor).

2. AI Acceleration & Edge Compute Optimization

- Integrated the deep-learning inference pipeline natively into a modern Reactive (Rx) streaming architecture. Engineered a custom zero-copy memory-management framework leveraging TensorRT, successfully maximizing sustained GPU utilization to 95% while strictly guaranteeing deterministic, safety-critical real-time latency limits.
- Engineered a dedicated scheduling module to actively drop stale sensor data based on strict latency budgets (e.g., 28 Hz for visual localization), enhancing both real-time performance and system issue reproducibility. Collaborated with hardware technicians using CPU/GPU thermal logs and power-consumption metrics to finalize the physical cooling redesign of the vehicle compute trunk.
- Orchestrated a strategic edge-compute offloading roadmap, leading technical integration with external hardware partners (Chimei). Architected a decentralized computing paradigm by strategically adopting edge ASICs to natively handle image compression and neural-network inference, permanently eliminating bulky frame grabbers and drastically alleviating central compute bottlenecks.

3. Technical Leadership & Infrastructure (DevOps)

- Directed technical execution for a cross-functional engineering team, scaling the unit from 4 to 10+ members. Formulated a comprehensive 4-year technical roadmap focused on resilient system scalability and continuous automated functional-safety validation.
- Orchestrated a massive 6-month ROS1 to ROS2 migration for a junior team. Defined strict SOPs and decomposed tasks (Message Interface, TF, Bag conversion). Established automated CI/CD pipelines on AWS and internal servers for bag-file playback, eradicating regression bugs during deployment.
- Engineered an automated Active-Learning data pipeline that executes offline bag-file playbacks against ML object detectors. Designed custom mismatch-trigger logic (Tracker vs. Detector discrepancies) to programmatically mine hard-negative edge cases, establishing a closed-loop data engine for continuous model refinement.

Education

National Yang Ming Chiao Tung University (NYCU / NCTU) — **Hsinchu, Taiwan**
M.S. in Electrical Engineering — *Sep 2016 – Jan 2019*

National Yang Ming Chiao Tung University (NYCU / NCTU) | **Hsinchu, Taiwan**
B.S. in Electrical Engineering | *Sep 2012 – Jun 2016*

Certifications & Patents

- **Automotive Functional Safety Professional (AFSP) – ISO 26262**, SGS TÜV Saar — Valid through Oct 2025.
- **3D Object Detection Method and System** (Sole Inventor), Dec 2024 — Taiwan App. 113150958 / Japan App. 2025-069543 (Pending).
- **Camera Calibration Method Based on Vehicle Localization** (Lead Inventor), Dec 2023 — US Patent 12,400,367; Taiwan I882582; Japan 2024-071947 (Allowed).
- **Safe Following Distance Estimation System and Method** (Co-Inventor), Dec 2021 — US Patent 11,891,063.

Publications

- **RMSeg-UDA: Unsupervised Domain Adaptation for Road Marking Segmentation**, ICRA 2025. Y.-C. Cai, H.-C. Hsiao, W.-C. Chiu, H.-Y. Lin, **C.-T. Chan**.
- **FuseRoad: Enhancing Lane Shape Prediction via Semantic Knowledge Integration**, IV 2025. H.-C. Hsiao, Y.-C. Cai, H.-Y. Lin, W.-C. Chiu, **C.-T. Chan**, C.-C. Wang.
- **MemPromptTSS: Persistent Prompt Memory for Iterative Multi-Granularity Time Series State Segmentation**, arXiv 2025. C. Chang, M.-C. Lo, **C.-T. Chan**, W.-C. Peng, T.-F. Chen.
- **TimeDRL: Disentangled Representation Learning for Multivariate Time-Series**, ICDE 2024. C. Chang, **C.-T. Chan**, W.-Y. Wang, W.-C. Peng, T.-F. Chen.
- **Self-Supervised Learning of Disentangled Representations for Multivariate Time-Series**, NeurIPS 2024 Workshop (Self-Supervised Learning — Theory and Practice). C. Chang, **C.-T. Chan**, W.-Y. Wang, W.-C. Peng, T.-F. Chen.
- **RLMD: A Dataset for Road Marking Segmentation**, ICCE-Taiwan 2023. H.-C. Hsiao, Y.-C. Cai, H.-Y. Lin, W.-C. Chiu, **C.-T. Chan**.

Core Competencies

- **Algorithm & Autonomy:** Visual-Inertial Localization, Sensor Fusion, Object Tracking (Fixed-Lag IMM-RAUKF, ICP)
- **AI Acceleration & Compute:** CUDA, TensorRT, Memory Management (Zero-copy), HW/SW Co-design
- **Infrastructure & MLOps:** C++, Python, ROS1/ROS2, AWS CI/CD, Nix, Active-Learning Data Pipelines
- **Leadership & Compliance:** ISO 26262 Functional Safety, Team Scaling, Roadmap Planning, Agile/SOP